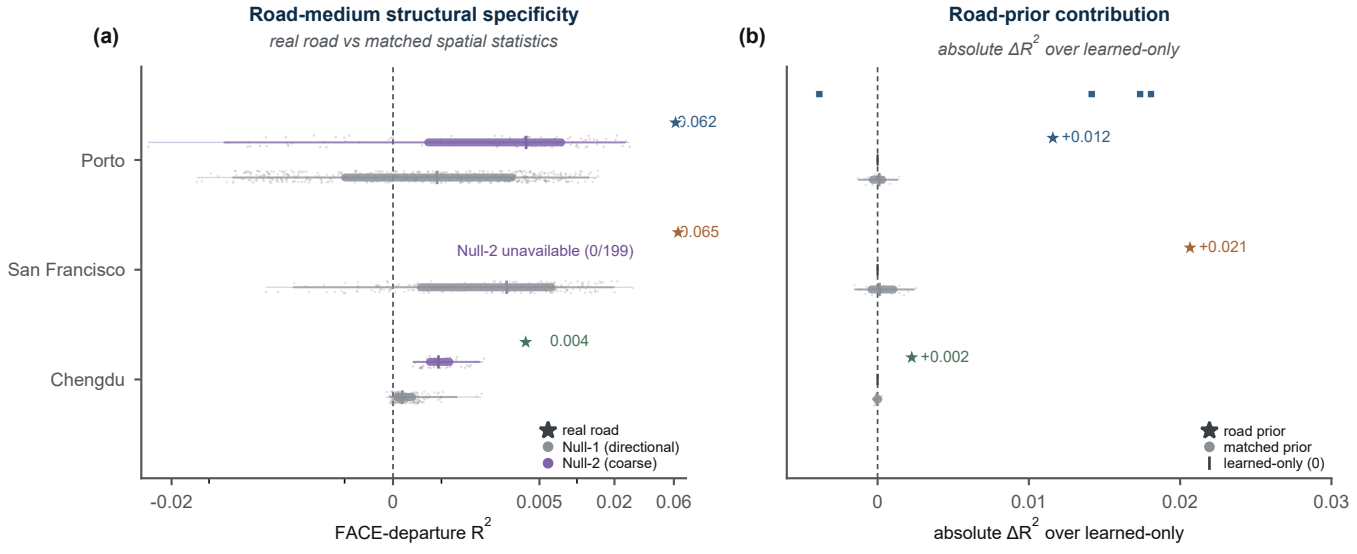

(a)
Road-medium structural specificity
real road vs matched spatial statistics
Porto
San Francisco
Chengdu
+0.062
+0.065
Null-2 unavailable (0/199)
0.004
real road
Null-1 (directional)
Null-2 (coarse)
-0.02
0
0.005
0.02
0.06
FACE-departure R²
(b)
Road-prior contribution
absolute ΔR² over learned-only
+0.012
+0.021
+0.002
road prior
matched prior
learned-only (0)
0
0.01
0.02
0.03
absolute ΔR² over learned-only